Samuel Silver (he/him/his)

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Research interests

• Algebraic/differential topology/geometry • Homotopy theory and higher category theory • Mathematical physics

Education

University of Massachusetts - Amherst Amherst, MA
BS in Mathematics (pure concentration). GPA: 4.0 Fall 2022 - Present, Expected graduation: Spring 2026
Relevant Coursework: Abstract Algebra I and II (611-612), Real and Complex Analysis (623, 621), Topology and Algebraic Topology (671-672), Lie Algebras (718), Representation Theory (717), Differential Geometry (703), Elliptic Curves (714), Algebraic Geometry (707), Higher Category Theory (690STJ), Quantum Field Theory (PHYSICS 811), all at the graduate level.

Waltham High School Waltham, MA
GPA: 5.1/5.3 weighted, SAT: 1580, ACT: 36 Fall 2018 - Spring 2022

Research Experience

Independent studies with Martina Rovelli Amherst, MA
Student Researcher Fall 2024 - Present

- Began studying the Morita construction for strict monoidal categories in Fall 2024, produced notes with proofs for the strict theory
- Collaborated closely with Martina and her students Arghan Dutta and Stefano Luneia to write a paper providing an explicit construction and verification of a 2-complicial set modeling the Morita $(\infty, 2)$-category of a general monoidal category
- Another paper in progress on upgrading the construction to a reduced $(\infty, 3)$-category given a braided monoidal category
- Currently turning the strict-case work into an honors thesis

Mathematics REU at the Hebrew University of Jerusalem Jerusalem, IL
Student Researcher Summer 2025

- Participated in the 2025 HUJI mathematics REU, working with Jake Solomon alongside two other students to search for further cases of the correspondence between descendent integrals on the moduli space $\overline{M}_{g,n,k}$ of marked genus-g Riemann surfaces with boundary and the moduli space $\overline{M}_{h,n}$ of marked genus-h curves

Mathematics REU with Jenia Tevelev Amherst, MA
Student Researcher Summer 2024

- Participated in a summer research program with Jenia Tevelev, in which I wrote a program in Julia to generate and check through a large class of potential counterexamples to the Fulton-Faber conjecture about generators for the cone of curves of $\overline{M}_{0,n}$, arising from matroids augmented with curves

UMass Amherst Mathematics REU Amherst, MA
Student Researcher Summer 2023

- Participated in the 2023 Math/Stats department REU program, in which our team wrote a paper characterizing the nilpotence variety $\mathcal{N}$ of the 2D superconformal Lie algebra $\mathfrak{sconf}(N)$ and the orbits of the action of $\mathfrak{sconf}_0(N)$ on $\mathcal{N}$
- This has applications to the physics of quantum field theory, particularly to theories of supersymmetry

Research Output

- Arghan Dutta, Stefano Luneia, Martina Rovelli, and Sam Silver, "The Morita $(\infty, 2)$-category of a monoidal category as a 2-complicial set", *submitted for publication*, preprint available at https://arxiv.org/abs/2509.21472

- Arghan Dutta, Stefano Luneia, Martina Rovelli, and Sam Silver, "The Morita $(\infty, 3)$-category of a strict braided monoidal category as a 3-complicial set", *in preparation*

- Sam Silver, "Notes on strict Morita categories", *in preparation for submission as an honors thesis to UMass Amherst CHC*, partial draft available at https://sam-silver.github.io/thesis_draft.pdf
- A database of curve-enhanced matroids and their status relative to the Fulton-Faber conjecture, available at https://sam-silver.github.io/CEM_DB.html
- Nikos Orginos, Nathaniel Reid, Sam Silver, "Adjoint actions of nilpotent elements in the 2D superconformal algebra $\mathfrak{sconf}(N)$", available at https://sam-silver.github.io/REU23.pdf

Talks

- "Classifying principal bundles over spheres using the clutching construction", 12 March 2025, Gauge theory reading seminar
- "What is a Feynman Diagram? A crash-course in perturbative QFT", 29 April 2025, Gauge theory reading seminar
- "What is the Morita bicategory of a monoidal category?", 8 October 2025, UMass Math Graduate Student Seminar
- "What is a hole?" (background-free talk defining the fundamental group to solve a puzzle), 16 October 2025, UMass Undergraduate Math Club

Leadership, Activities, & Employment

UMass Recreational Math Club — Amherst, MA
President — since Fall 2025

- Planning and running weekly meetings, which typically consist of pizza and a new problem set for people to solve together
- Also assisting with budgeting and advertisement for the club
- 40+ regular members

UMass Mathematics and Statistics Department — Amherst, MA
Grader — since Fall 2025

- Grading weekly homeworks and quizzes for MATH 545: Advanced Linear Algebra

City of Waltham Recreation Department — Waltham, MA
Assistant park ranger — since Summer 2022

- Working seasonally (whenever possible) at Prospect Hill Park in Waltham, MA to keep the park tidy and safe, install trail signs, etc. Avid enjoyer of the outdoors.

Boy Scouts of America, Troop 250 Waltham — Waltham, MA
Patrol Leader, Quartermaster, Eagle Scout — October 2020 - June 2022

- Active Boy Scout since 1st grade
- Lead and managed activities and planning for my patrol
- Closely involved in planning and leadership for the troop as a whole, as well as organization of supplies
- Became an Eagle Scout in 2022
  - Culmination of efforts as a Boy Scout
  - Prepared, gathered materials and assistance, and led a team of volunteers to install 20 new signposts in Prospect Hill Park in Waltham, MA
  - Significantly developed leadership, organizational, and project management skills

Skills & Interests

Technical: Experience with Python/Sage and Julia scripting, and an interest in cybersecurity as a hobby. Involved from time to time in the UMass Cybersecurity Club, particularly with cryptography.
Language: English (native), French (conversational), Russian (conversational)